AN INTELLIGENT FRAMEWORK FOR MODELING AND SIMULATING STARTUP BUSINESS PROCESS

A PROJECT REPORT

Submitted by

KAVIN KIRTHIK RP	**811722104071**
NIKHAAL AHAMED A K	**811722104101**
RAJESH R	**811722104117**

in partial fulfillment of the requirements for the award degree of

Bachelor of Engineering

20CS8504 – PROJECT WORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM – 621112

APRIL 2026

AN INTELLIGENT FRAMEWORK FOR MODELING AND SIMULATING STARTUP BUSINESS PROCESS

A PROJECT REPORT

Submitted by

KAVIN KIRTHIK RP	**811722104071**
NIKHAAL AHAMED A K	**811722104101**
RAJESH R	**811722104117**

in partial fulfillment of the requirements for the award degree of

Bachelor of Engineering

20CS8504 – PROJECT WORK

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY

(AUTONOMOUS)

SAMAYAPURAM – 621112

APRIL 2026

K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY
(AUTONOMOUS)
SAMAYAPURAM - 621112

BONAFIDE CERTIFICATE

The work embodied in the present project report entitled "**AN INTELLIGENT FRAMEWORK FOR MODELING AND SIMULATING STARTUP BUSINESS PROCESS**" has been carried out by the students **KAVIN KIRTHIK RP , NIKHAAL AHAMED A K , RAJESH R** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

Date of Viva Voce: ……

Mrs. S. GAYATHRI ,M.E	**Mr. R. RAJAVARMAN, M.E.,(Ph.D.,)**
SUPERVISOR	HEAD OF THE DEPARTMENT
Assistant Professor	Assistant Professor
Department of CSE	Department of CSE
K. Ramakrishnan College of Technology	K.Ramakrishnan College of Technology
(Autonomous)	(Autonomous)
Samayapuram-621 112	Samayapuram-621 112

INTERNAL EXAMINER **EXTERNAL EXAMNIER**

ABSTRACT

The rapid growth of startups and entrepreneurial activities has increased the need for practical learning platforms that help individuals understand real-world business processes. Traditional learning methods mainly focus on theoretical knowledge and do not provide sufficient exposure to real-time decision-making and business challenges. The system is developed using the Unity game engine and is designed to guide users through different stages of a startup lifecycle, including domain selection, company registration, product development, and product deployment. Each stage acts as a level where players make decisions related to resource allocation, marketing strategies, and financial planning. To enhance decision-making and learning, the system integrates Artificial Intelligence using Large Language Models through the Ollama server. A rule-based decision system evaluates user choices, while adaptive gameplay mechanisms dynamically adjust market conditions, competition, and risks based on player performance. The system's interactive design, real-time feedback, and adaptive features make it suitable for applications in education, training, and business simulation environments.

Keywords: Startup Simulation, Game-Based Learning, Artificial Intelligence, Unity Game Engine, Business Simulation, Adaptive Gameplay

ACKNOWLEDGEMENT

We thank our **Dr. N. Vasudevan**, Principal, for his valuable suggestions and support during the course of my research work.

We thank our **Mr. R. Rajavarman,** Head of the Department, Assistant Professor (Sr. Grade), Department of Computer Science and Engineering, for his valuable suggestions and support during the course of my research work.

We wish to record my deep sense of gratitude and profound thanks to my Guide **Mrs. S. Gayathri,** Assistant Professor, Department of Computer Science and Engineering for her keen interest, inspiring guidance, constant encouragement with my work during all stages, to bring this thesis into fruition.

We are extremely indebted to our project coordinator **Mrs. M. Mathumathi,** Assistant Professor, Department of Computer Science and Engineering, for her valuable suggestions and support during the course of my research work.

We also thank the faculty and non-teaching staff members of the Computer Science and Engineering, K. Ramakrishnan College of Technology, Samayapuram, for their valuable support throughout the course of my research work.

Finally, we thank our parents, friends and our well wishes for their kind support.

SIGNATURE

_____

_____

_____

TABLE OF CONTENTS

CHAPTER	TITLE	PAGE NO.
	ABSTRACT	**iii**
	LIST OF FIGURES	**vii**
	LIST OF ABBREVIATIONS	**viii**
1	**INTRODUCTION**	**1**
	1.1 DESCRIPTION	1
	1.2 STARTUP BUSINESS PROCESS SIMULATION	2
	1.2.1 Game Development	3
	1.2.2 Artifical intelligence	4
	1.2.3 Business Simulation	4
	1.2.4 Intelligent Decision	5
	1.3 SCOPE OF THE PROJECT	5
2	**LITERATURE SURVEY**	**6**
3	**EXISTING SYSTEM**	**16**
	3.1 EXISTING SYSTEM	16
	3.2 EXISTING SYSTEM ARCHITECTURE	17
4	**PROBLEMS IDENTIFIED**	**18**
5	**PROPOSED SYSTEM**	**21**
	5.1 PROPOSED SYSTEM	21
	5.2 SYSTEM ARCHITECTURE	22
6	**SYSTEM REQUIREMENTS**	**23**
	6.1 HARDWARE REQUIREMENTS	23
	6.2 SOFTWARE REQUIREMENTS	23
7	**SYSTEM IMPLEMENTATIONS**	**24**
	7.1 LIST OF MODULES	24
	7.2 MODULES DESCRIPTION	24
	7.2.1 Player Interaction Module	24

	7.2.2 Domain Selection module	25
	7.2.3 Company Registration module	25
	7.2.4 Product Development module	26
	7.2.5 Product Deployment module	26
8	**SYSTEM TESTING**	**27**
	8.1 SYSTEM TESTING	27
	8.2 UNIT TESTING	27
	8.3 INTEGRATION TESTING	28
	8.4 PERFORMANCE TESTING	28
	8.5 SECURITY TESTING	29
	8.6 USABILITY TESTING	29
9	**RESULTS AND DISCUSSION**	**30**
10	**CONCLUSION AND FUTURE ENHANCEMENTS**	**33**
	10.1 CONCLUSION	33
	10.2 FUTURE ENHANCEMENTS	34
	APPENDIX A - SOURCE CODE	**36**
	APPENDIX B - SCREENSHOTS	**45**
	REFERENCES	**48**

LIST OF FIGURES

FIGURE NO.	FIGURE NAME	PAGE NO.
3.1	Existing System diagram	17
5.1	Proposed System diagram	22
B.1	Game Map	45
B.2	Registration section	45
B.3	3d map	46
B.4	Chat bot interface	46
B.5	Hiring Section	47

LIST OF ABBREVIATIONS

ABBREVIATION		FULL FORM
AI	-	Artificial Intelligence
LLM	-	Large Language Model
NLP	-	Natural Language Processing
UI	-	User Interface
API	-	Application Programming Interface
UX	-	User Experience
GPU	-	Graphics Processing Unit
CPU	-	Central Processing Unit
IPC	-	Inter-process Communication
JSON	-	JavaScript Object Notation
DB	-	Database
IDE	-	Integrated Development Environment